

The Grand Synthesis: Network to Link Layer

Expanded Review: Data Plane, Control Plane, and LANs

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Agenda: The Big Picture

- **Part 1: The Network Layer - Data Plane (Ch. 4)**
- **Part 2: The Network Layer - Control Plane (Ch. 5)**
- **Part 3: Transition to the Link Layer (Ch. 6)**
- **Part 4: Synthesis and Review**

The Network Layer: Two Interacting Planes (Section 4.1)

The Network Layer is fundamentally divided into two distinct components:

- **The Data Plane (Forwarding):** The local, per-router action. It moves a datagram from a router's input link to the appropriate output link. It happens in nanoseconds and is implemented in hardware.
- **The Control Plane (Routing):** The network-wide logic. It determines the end-to-end path from source to destination. It happens in milliseconds or seconds and is implemented in software.

Router Architecture: Input Ports (Section 4.2)

What happens the moment a physical signal enters the router?

- **Line Termination:** Receives the physical, analog signal and converts it to digital bits.
- **Link Layer Processing:** Strips away the Ethernet frame to extract the IP datagram.
- **Lookup and Forwarding:** The most critical step. The input port consults the forwarding table to find the correct output port based on the destination IP address.
- **Queuing:** If datagrams arrive faster than the internal switching fabric can move them, they queue up here.

Router Architecture: The Switching Fabric (Section 4.2)

The physical heart of the router that connects inputs to outputs.

- **Memory:** The oldest method. The input port copies the packet to the router's main memory, the CPU processes it, and copies it to the output. Slow and limited by RAM speed.
- **Bus:** Packets travel over a shared internal bus. Faster, but only one packet can cross the bus at a time (bus contention).
- **Crossbar / Interconnection Network:** A grid of parallel buses. Allows multiple packets to be forwarded to different output ports simultaneously. Used in all high-end modern routers.

Router Architecture: Output Ports & Queuing (Section 4.2)

- **Buffering:** Output ports must store packets if the switching fabric delivers them faster than the physical transmission rate of the outgoing link.
- **Packet Loss:** If the output queue fills up completely, newly arriving packets are dropped. This is the primary cause of packet loss on the Internet.
- **Scheduling:** The router decides which queued packet to send next. Standard is First-In-First-Out (FIFO), but Quality of Service (QoS) can prioritize VoIP over email traffic.

IPv4 Addressing and CIDR (Section 4.3)

- **Structure:** An IPv4 address is 32 bits long, typically written in dotted-decimal notation.
- **Network vs. Host:** The address is split. The first part identifies the subnet; the second part identifies the specific host.
- **CIDR (Classless Inter-Domain Routing):** The modern standard. Represented as a.b.c.d/x, where x is the number of bits in the subnet portion.
- **Why CIDR?** It prevents the massive waste of IP addresses caused by the rigid, outdated Class A, B, and C system.

DHCP: The Plug-and-Play Protocol (Section 4.3)

How does a device securely and automatically obtain an IP address?

- ➊ **Discover:** The arriving client broadcasts a DHCP Discover message to find a server.
- ➋ **Offer:** The DHCP server responds with an offered IP address and a lease time.
- ➌ **Request:** The client formally requests the offered IP.
- ➍ **ACK:** The server acknowledges, confirming the client can use the IP.

Bonus: DHCP also hands out the Subnet Mask, Default Gateway router IP, and DNS server IP.

NAT: Network Address Translation (Section 4.3)

The IPv4 address space (4.3 billion) was exhausted years ago. NAT saved the Internet.

- **Private Realms:** Inside a home or office, all devices use private IP addresses (e.g., 192.168.1.x). These cannot be routed on the global Internet.
- **The Translation Table:** When a packet leaves the house, the NAT router replaces the internal Private IP with its single, public WAN IP.
- **Port Mapping:** To remember which internal device sent the request, NAT assigns a unique source Port Number to each connection.

The Control Plane: Two Algorithm Types (Section 5.2)

To build the forwarding tables, routers run mathematical algorithms.

- **Link-State (LS):** Requires a global map. Every router broadcasts its link costs to every other router. Every router builds the same Adjacency Matrix and runs Dijkstra's algorithm.
- **Distance-Vector (DV):** Decentralized. No router has the full map. A router only knows the cost to its physical neighbors and iteratively shares its own distance estimates.

Distance-Vector: The Bellman-Ford Equation (Section 5.2)

Unlike Dijkstra's, the DV algorithm relies on the Bellman-Ford equation:

$$d_x(y) = \min_v \{c(x, v) + d_v(y)\}$$

- $d_x(y)$ is the least-cost path from node x to node y .
- Node x looks at all its immediate neighbors (v).
- It adds the direct cost to that neighbor $c(x, v)$ to the neighbor's advertised cost to reach the destination $d_v(y)$.
- It simply picks the minimum value.

Scaling the Internet: Autonomous Systems (Section 5.3)

Both LS and DV algorithms break down at a global scale due to memory, CPU, and broadcast limits ($\mathcal{O}(n^2)$ complexity).

- **Partitioning:** The Internet is divided into Autonomous Systems (AS), also known as domains.
- **Administrative Control:** An AS consists of a group of routers under the same administrative control (e.g., an ISP or a large university).
- **Intra-AS Routing:** Routing algorithms only run *inside* the AS, vastly reducing the math required.

Intra-AS Routing: OSPF (Section 5.3)

Open Shortest Path First (OSPF) is the standard for routing inside an AS.

- **Protocol Basics:** It is a Link-State protocol that uses Dijkstra's algorithm.
- **Hierarchical OSPF:** For massive Autonomous Systems, OSPF can be split further into Local Areas and a central Backbone Area to contain broadcast traffic.
- **Area Border Routers:** These specialized routers connect the local areas to the backbone, summarizing distances so local routers don't need the full backbone map.

Border Gateway Protocol (BGP) routes traffic *between* different Autonomous Systems.

- **Path Vector:** BGP routers advertise entire paths (sequences of AS numbers) rather than just link costs.
- **eBGP vs iBGP:**
 - **eBGP** (External): The connection between a gateway router in AS1 and a gateway router in AS2.
 - **iBGP** (Internal): The connection used to broadcast that external knowledge to all other routers *inside* AS1.

BGP Attributes: AS-PATH and NEXT-HOP (Section 5.4)

- **AS-PATH Attribute:** Lists the Autonomous Systems the route advertisement has traversed.
 - *Purpose:* Loop prevention. If a BGP router receives an advertisement that includes its own AS number, it drops the route.
- **NEXT-HOP Attribute:** The specific IP address of the router interface that begins the AS-PATH.
 - *Purpose:* It tells the internal routers (via iBGP) the exact IP address of the physical gateway they must target to leave the AS.

BGP Route Selection (Section 5.4)

When a BGP router receives multiple paths to the same destination, it applies these rules in order:

- ① **Local Preference:** Highest preference wins. This is manually set by administrators for business/policy reasons.
- ② **Shortest AS-PATH:** If tied, pick the route crossing the fewest networks.
- ③ **Hot Potato Routing:** If tied, pick the path with the lowest *internal* cost to reach the NEXT-HOP gateway. Get it out of the network fast!
- ④ **BGP Identifiers:** If still tied, use router IDs.

SDN: Software-Defined Networking (Section 5.5)

- **The Distributed Era:** Historically, every router ran its own control plane software (OSPF/BGP) and calculated its own tables.
- **The SDN Era:** The control plane is physically separated.
- **How it works:** A remote, logically centralized **SDN Controller** server calculates all the routes. It then pushes the simple forwarding tables down to the physical switches using protocols like OpenFlow.
- **Advantage:** Allows for programmatic, automated traffic engineering across entire data centers.

Transition: Down to the Link Layer (Section 6.1)

The Control Plane found the path. The Data Plane looked up the port. Now we must physically cross the wire.

- The Network Layer does not handle physical transmission. It passes the datagram down to the **Link Layer**.
- **Core Responsibility:** Node-to-node delivery. Moving the datagram from one node to the physically adjacent node over a single link.
- **Framing:** The Link Layer wraps the IP datagram in a Link-Layer **Frame** (adding MAC addresses and error detection).

Error Detection: 2D Parity and CRC (Section 6.2)

Physical signals degrade. The Link Layer uses hardware to detect flipped bits.

- **Two-Dimensional Parity:** Bits are arranged in a matrix with parity bits for rows and columns. This allows the receiver to pinpoint and correct a single bit error (Forward Error Correction).
- **Cyclic Redundancy Check (CRC):** The gold standard. Uses polynomial modulo-2 arithmetic. The sender calculates an R -bit trailer so the entire frame is cleanly divisible by a generator G . The receiver divides; any remainder means corruption.

The Multiple Access Problem (Section 6.3)

- **Broadcast Links:** Multiple nodes share a single communication channel (like a room of people, or a single Wi-Fi frequency).
- **Collisions:** If two nodes transmit simultaneously, their electromagnetic signals combine and interfere. Both frames are completely lost.
- **MAC Protocols:** We need Medium Access Control protocols to coordinate who gets to speak and when.

CSMA/CD: Ethernet's Solution (Section 6.3.2)

Carrier Sense Multiple Access with Collision Detection.

- **Carrier Sense:** Listen before speaking. If the wire is active, wait.
- **Collision Detection:** Stop if interrupted. If a collision happens while transmitting, abort immediately to save bandwidth.
- **Binary Exponential Backoff:** After a collision, nodes pick a random wait time from a specific set. After 1 collision, choose from $\{0, 1\}$. After 2 collisions, choose from $\{0, 1, 2, 3\}$. The range doubles exponentially.

MAC Addresses (Section 6.4.1)

The Link Layer does not use IP addresses.

- **Physical Addresses:** We use Media Access Control (MAC) addresses.
- **Format:** 48 bits long, represented as 6 hexadecimal pairs (e.g., 1A:2B:3C:4D:5E:6F).
- **Permanent:** Burned into the Network Interface Controller (NIC) at the factory. It never changes, unlike an IP address which changes based on your location.

ARP: Address Resolution Protocol (Section 6.4.1)

If your computer knows the destination IP, how does it find the MAC address to build the frame?

- **ARP Request:** The sender broadcasts a query to the special broadcast MAC address `FF:FF:FF:FF:FF:FF`. All devices on the local subnet receive it.
- **ARP Reply:** The specific device holding that IP address replies with its MAC address.
- **ARP Table:** The sender caches this IP-to-MAC mapping locally so it doesn't have to broadcast every time.

Ethernet Frame Structure (Section 6.4.2)

- **Preamble (8 bytes):** A specific pattern of 1s and 0s used to wake up the receiving hardware and synchronize internal clocks.
- **Destination MAC (6 bytes):** Where the frame is going locally.
- **Source MAC (6 bytes):** Where the frame came from locally.
- **Type (2 bytes):** Identifies the upper-layer protocol (usually IPv4 or ARP).
- **Payload (46-1500 bytes):** The IP Datagram.
- **CRC (4 bytes):** The error-checking trailer.

Link-Layer Switches (Section 6.4.3)

- **Layer 2 Devices:** Switches do not care about IP addresses. They forward frames based strictly on Destination MAC addresses.
- **Transparent:** Hosts don't explicitly address frames to the switch; they address them to the final destination.
- **No Collisions:** Switches have dedicated, full-duplex links to every host and buffer incoming frames. Therefore, modern switched Ethernet has zero collisions and doesn't use CSMA/CD!

Switch Tables and Self-Learning (Section 6.4.3)

A switch builds its forwarding table automatically.

- 1 **Learning:** When a frame arrives on Port 1, the switch reads the *Source MAC*. It now knows that MAC is reachable via Port 1 and records it in its table.
- 2 **Forwarding:** It looks at the *Destination MAC*. If it exists in the table, it forwards it out the specific port.
- 3 **Flooding:** If the Destination MAC is unknown, the switch broadcasts the frame out of *all* ports (except the incoming port).

VLANs: Virtual LANs (Section 6.4.4)

- **The Problem:** A large flat switched network means broadcast traffic (like ARP) hits every single host. This wastes bandwidth and poses security risks.
- **The Solution:** Port-based VLANs allow admins to logically divide a physical switch into multiple isolated broadcast domains via software.
- **Trunking (802.1Q):** To connect VLANs across different physical switches, a trunk port carries all traffic, adding a 4-byte VLAN tag to the Ethernet frame so the receiving switch knows where it belongs.

A Day in the Life: Getting Connected (Section 6.7)

Let's put it all together. You connect your laptop to campus Wi-Fi.

- 1 You have no IP. Your laptop broadcasts a **DHCP Discover** message encapsulated in UDP, IP, and Ethernet.
- 2 The **DHCP Server** replies, giving you an IP address, Subnet Mask, Gateway Router IP, and DNS Server IP.
- 3 You want to visit `google.com`. You need its IP address. You formulate a **DNS Query**.
- 4 To send the DNS query, you must reach the Gateway Router. You broadcast an **ARP Request** to find the router's MAC address.

A Day in the Life: Fetching the Data (Section 6.7)

- 5 The Gateway Router replies with its MAC via **ARP Reply**.
- 6 You encapsulate the DNS query into an Ethernet frame targeted at the Gateway's MAC address.
- 7 The Router uses the Control Plane (**OSPF** and **BGP**) to forward your DNS query across the Internet via the Data Plane.
- 8 DNS replies with Google's IP address.
- 9 You initiate a 3-way **TCP Handshake** and send the **HTTP GET** request to load the page!